Midterm Sample 1: Solutions

ECON 441: Introduction to Mathematical Economics — Instructor: Div Bhagia

1. (4.5 pts) Answer the following questions

   (a) (1 pt) Consider two sets A and B, where A is the set of all odd real numbers and B is the set of all even real numbers. What is the union of A and B?

$$A \cup B = \text{Set of all integers}$$

   (b) (1 pt) Expand the following summation expression:

$$\sum_{j=1}^{2}\sum_{i=1}^{2} X_iY_j = \sum_{j=1}^{2}\left(\sum_{i=1}^{2} X_iY_j\right) = \sum_{j=1}^{2}\left(X_1Y_j + X_2Y_j\right) = X_1Y_1 + X_2Y_1 + X_1Y_2 + X_2Y_2$$

   (c) (1pt) Does the inverse of the function $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ exist? If yes, find the inverse.

   No, since this function is non-monotonic, the inverse does not exist.

   (d) (1.5 pts) Find A^TA where

$$A = \begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

   Remember: A^T is the transpose of A.

$$A^TA = \begin{bmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{bmatrix}$$

$$A^TA = \begin{bmatrix} \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) & \left(\frac{1}{\sqrt{2}} \times -\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) \\ \left(-\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) & \left(-\frac{1}{\sqrt{2}} \times -\frac{1}{\sqrt{2}}\right) + \left(\frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}}\right) \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

2. (8 pts) Two types of cars, gasoline (g) and electric (e), are available in the market. Denote the price of gasoline cars by p_g and the price of electric cars by p_e. Similarly, denote the quantity of gasoline cars as q_g and electric cars as q_e. The supply and demand equations for these cars are as follows:

Supply Equations:

$$q_e^s = 25 + 0.3p_e$$
$$q_g^s = 50 + 0.3p_g$$

Demand Equations:

$$q_e^d = 100 - 0.5p_e + 0.2p_g$$
$$q_g^d = 150 + 0.2p_e - 0.5p_g$$

Note: At equilibrium, supply is equal to demand: $q_e^s = q_e^d = q_e$ and $q_g^s = q_g^d = q_g$.

(a) (2 pts) Write down the <u>four</u> equations that must hold in equilibrium. Rearrange each equation so that all terms with variables are on one side and all constants are on the other. What are the four unknown variables?

$$q_e - 0.3p_e = 25 \tag{1}$$
$$q_g - 0.3p_g = 50 \tag{2}$$
$$q_e + 0.5p_e - 0.2p_g = 100 \tag{3}$$
$$q_g - 0.2p_e + 0.5p_g = 150 \tag{4}$$

Four unknown variables: q_e, p_e, q_g, p_g.

(b) (2 pts) Express this system of equations in matrix format as $Ax = b$. Clearly specify what A, x, and b are.

$$\underbrace{\begin{pmatrix} 1 & -0.3 & 0 & 0 \\ 0 & 0 & 1 & -0.3 \\ 1 & 0.5 & 0 & -0.2 \\ 0 & -0.2 & 1 & 0.5 \end{pmatrix}}_{A} \underbrace{\begin{pmatrix} q_e \\ p_e \\ q_g \\ p_g \end{pmatrix}}_{x} = \underbrace{\begin{pmatrix} 25 \\ 50 \\ 100 \\ 150 \end{pmatrix}}_{b}$$

(c) (2 pts) What is the necessary and sufficient condition for a unique solution for this system of equations to exist?

Necessary condition: A must be a square matrix, meaning the number of equations is equal to the number of variables; this condition is met here.

Sufficient condition: A must be non-singular (i.e., $|A| \neq 0$), indicating that the rows of A are linearly independent or, equivalently, that each equation is linearly independent.

(d) (2 pts) How would you solve this system of equations using tools from matrix algebra? There's no need to actually solve it; just explain the steps involved.

To solve this system of equations, we can start by noting that pre-multiplying $Ax = b$ by A^{-1} gives us $x = A^{-1}b$. So we can find the inverse of the matrix A and multiply it with b to find x.